

BARANI C

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🐙 <https://github.com/Barospooky>

SUMMARY

Detail-oriented and passionate Electronics and Communication Engineering graduate with a strong software and hardware foundation. Skilled in MATLAB, Full stack development, Machine Learning. Enthusiastic about applying engineering principles to solve real-world challenges.

EDUCATION

Bachelor of Engineering in Electronics and Communication Engineering,

K.S.R. College of Engineering - 2021 – 2025.

CGPA – 8.6

Higher Secondary School,

Government Higher Sec School, Kozhikkalnatham – 2021

87.67%

SKILLS

- **Software** - MATLAB, Simulink, Microsoft Excel, Power Bi
- **Programming** - C,C++,Python, Javascript, HTML/CSS ,Typescript
- **Frontend** - React.js, Tailwind CSS, Bootstrap
- **Backend** - Node.js, Express.js, Sequelize ORM, PostgreSQL, MySQL, MongoDB
- **Developer Tools** - Git, Github , Docker, Postman

INTERNSHIPS

INDIAN SPACE RESEARCH ORGANIZATION (SAC)

2024

@ Ahmedabad, Gujarat.

Research Intern:

- Worked on a project titled "Simulation of Fine Pointing System using FSM and Quad Detector" for PAT subsystem in an optical Ground station.
- Observed the Gaussian nature of laser spot and calculated its position when moves away from the center of the quad detector using **MATLAB**.
- Simulated a fine pointing system using **SIMULINK** and Compared various centroid algorithms which uses voltage as a parameter such as sum difference algorithm, diagonal algorithm and difference over sum algorithm to find the centroid of laser (beacon) spot and founded the optimised one using **EXCEL** graphs.

PROJECTS

Asset Management

- Developed a full-stack Asset Management web application using Node.js, Express, PostgreSQL, and Sequelize ORM.
- Designed and implemented Employee, Asset, and Asset Category Masters with add/edit/view and search/filter functionality.

- Built asset lifecycle management features including issue, return (with reasons), scrap, and complete asset history tracking and created a stock dashboard providing branch-wise totals and overall asset valuation.
- Used Jade templating engine, Bootstrap, CSS, and DataTables.net to build an interactive and responsive UI.

Zomato Clone

- Built a Zomato-like food ordering web app with MongoDB, Express.js, React, and Node.js (MERN).
- Implemented user authentication with JWT tokens, enabling secure login and session handling and designed category-based food browsing with dynamic rendering of menu items.
- Developed Add to Cart & Checkout flow with real-time total price calculation and a popup summary modal and followed MVC architecture and integrated RESTful APIs for smooth frontend–backend communication.

Air Quality Analysis, Handwritten Digit Recognition using Machine learning in python

- Built an Air Quality Analysis tool leveraging IoT sensors and Python data analytics to monitor real-time pollution levels, generating actionable insights via visualization (Matplotlib).
- Designed a Handwritten Digit Recognition model using machine learning on the MNIST dataset, achieving over 98% accuracy for automated digit classification.

EXTRA-CURRICULAR ACTIVITIES & ACHIEVEMENTS

CERTIFICATION:

- Completed an advanced level training course on "Power Bi" in collaboration with I-BACUS TECH.

PAPER PRESENTATION:

- Topic on "Cloud computing" at karpagam college of engineering (I provided examples of well-known cloud services while clearly defining the idea and making it accessible to everyone)
- Topic on "Feature of Ultracapacitor in Automobiles" at Bannari Amman Institute Of Technology(gave a presentation outlining the benefits of using ultracapacitors as energy storage options in cars.

WORKSHOP:

- Topic on "Electric And Hybrid Vechicle" at **IIT MADRAS** (Acquired knowledge about the integration of electric power trains, battery systems, and eco-friendly transportation solutions)
- Topic on "Artificial Intelligence with Machine Learning" at **IISC Bangalore** (Learned to use AI and ML for smart decisions, creative solutions).

PERSONAL INFO

Date of Birth: 27-03-2003

Communication Address: 5/228, Selanthanthoppu,Tiruchengode,Namakkal,Tamil Nadu – 637209.

Languages Known: Tamil, English, Hindi, Marathi

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